

## 3740. Minimum Distance Between Three Equal Elements I

Solved 

Easy

 Topics

 Companies

 Hint

You are given an integer array `nums`.

A tuple `(i, j, k)` of 3 **distinct** indices is **good** if `nums[i] == nums[j] == nums[k]`.

The **distance** of a **good** tuple is `abs(i - j) + abs(j - k) + abs(k - i)`, where `abs(x)` denotes the **absolute value** of `x`.

Return an integer denoting the **minimum** possible **distance** of a **good** tuple. If no **good** tuples exist, return `-1`.

### Example 1:

**Input:** `nums = [1,2,1,1,3]`

**Output:** 6

**Explanation:**

The minimum distance is achieved by the good tuple `(0, 2, 3)`.

`(0, 2, 3)` is a good tuple because `nums[0] == nums[2] == nums[3] == 1`. Its distance is `abs(0 - 2) + abs(2 - 3) + abs(3 - 0) = 2 + 1 + 3 = 6`.

### Example 2:

**Input:** `nums = [1,1,2,3,2,1,2]`

**Output:** 8

**Explanation:**

The minimum distance is achieved by the good tuple `(2, 4, 6)`.

`(2, 4, 6)` is a good tuple because `nums[2] == nums[4] == nums[6] == 2`. Its distance is `abs(2 - 4) + abs(4 - 6) + abs(6 - 2) = 2 + 2 + 4 = 8`.

### Example 3:

**Input:** `nums = [1]`

**Output:** `-1`

#### Explanation:

There are no good tuples. Therefore, the answer is -1.

#### Constraints:

- `1 <= n == nums.length <= 100`
- `1 <= nums[i] <= n`

## Python:

class Solution:

def minimumDistance(self, nums: List[int]) -> int:

n = len(nums)

last2 = [0] \* n

res = 200

for i in range(n):

val, pos = nums[i] - 1, i + 1

pack = last2[val]

old, cur = pack & 255, pack >> 8

last2[val] = cur | (pos << 8)

if old:

res = min(res, (pos - old) << 1)

return -(res == 200) | res

## JavaScript:

```
const minimumDistance = nums => {  
  const len = nums.length;
```

```

const last2 = Array(len).fill(0);
let res = 200;

for (let i = 0; i < len; i++) {
  const val = nums[i] - 1;
  const pos = i + 1, pack = last2[val];
  const old = pack & 255, cur = pack >> 8;

  last2[val] = cur | (pos << 8);

  if (old)
    res = Math.min(res, (pos - old) << 1);
}

return -(res === 200) | res;
};

```

## Java:

```

class Solution {
  public int minimumDistance(int[] nums) {
    int len = nums.length;
    int[] last2 = new int[len];
    int res = 200;

    for (int i = 0; i < len; i++) {
      int val = nums[i] - 1;
      int pos = i + 1, pack = last2[val];
      int old = pack & 255, cur = pack >> 8;

      last2[val] = cur | (pos << 8);

      if (old > 0)
        res = Math.min(res, (pos - old) << 1);
    }

    return res == 200 ? -1 : res;
  }
}

```